



University of Wisconsin
SCHOOL OF MEDICINE
AND PUBLIC HEALTH

BMI/STAT-768

Statistical Methods for Medical Image Analysis and High Dimensional Data

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<https://github.com/lapcebeltrami/BMI768>

Diffusion

Unit Objectives

- 1) Understand diffusion equations
- 2) Understand how to solve them analytically
- 3) Understand how to solve them numerically

Required reading for diffusion on manifold

Chung et al. arXiv:2203.06665

Laplace Operator in Euclidean space

The Laplacian measures how a function differs from the average of its neighboring values.

It's the sum of second derivatives — how “curved” the function is at a point!



1 Definition (continuous)

In 2D Euclidean space $(x, y) \in \mathbb{R}^2$,

$$\Delta f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

The Laplacian tells us how much f at a point differs from its local average.

2 Finite difference approximation

On a grid with spacing $\delta x = \delta y = 1$, we approximate second derivatives by central differences:

$$\frac{\partial^2 f}{\partial x^2}(x, y) \approx f(x+1, y) - 2f(x, y) + f(x-1, y)$$

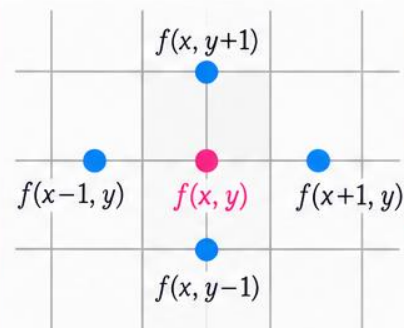
$$\frac{\partial^2 f}{\partial y^2}(x, y) \approx f(x, y+1) - 2f(x, y) + f(x, y-1)$$

Therefore,

$$\Delta f(x, y) \approx f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$

3 5-point stencil

The discrete Laplacian uses the center point and its four neighbors (up, down, left, right).



It's like comparing the center value to the average of its neighbors!



4 Matrix form

We can write the discrete Laplacian as

$$\Delta f = \sum_{i,j=-1,0,1} w_{ij} f(x_i, y_j)$$

with weights

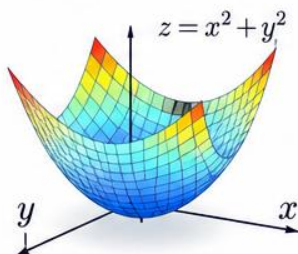
$$(w_{ij}) = \begin{pmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

This is called the 5-point Laplacian stencil.

5 Example

If $f(x, y) = x^2 + y^2$, then

$$\Delta f = 2 + 2 = 4.$$



A larger Laplacian means the function is more curved (locally).



6 Higher dimensions

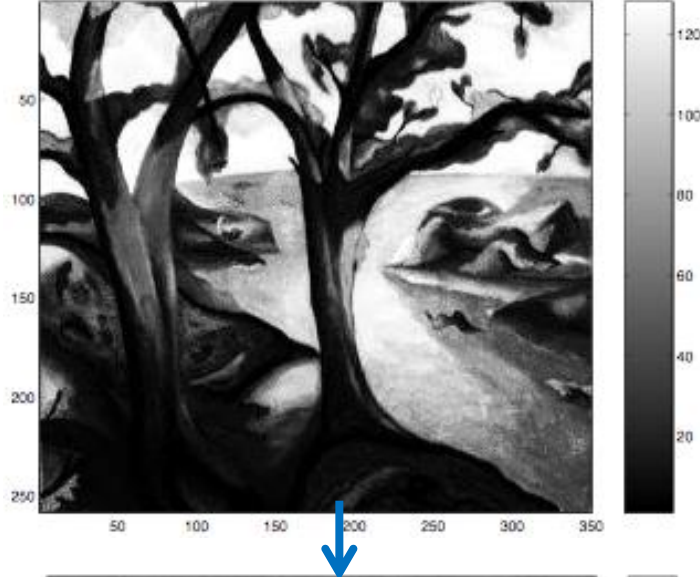
In $\mathbb{R}^n$, the Laplacian is the sum of second partial derivatives:

$$\Delta f = \sum_{k=1}^n \frac{\partial^2 f}{\partial x_k^2}$$

On a regular grid with unit spacing, each point has $2n$ neighbors and the center weight is $-2n$.

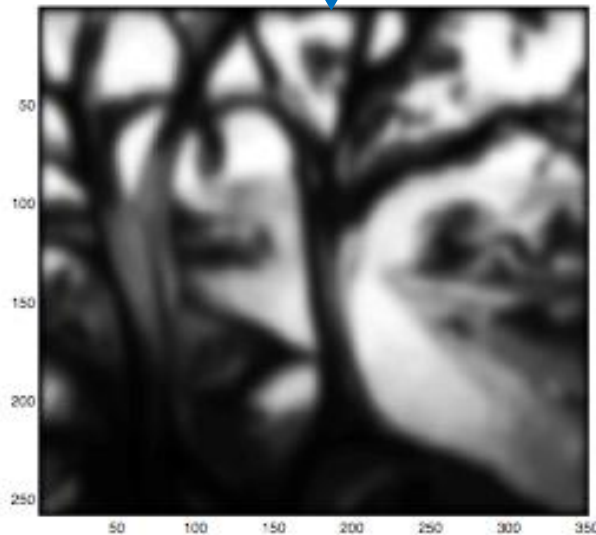
The Laplacian captures how a function spreads out or deviates from its local average.

Edge detection using Laplacian

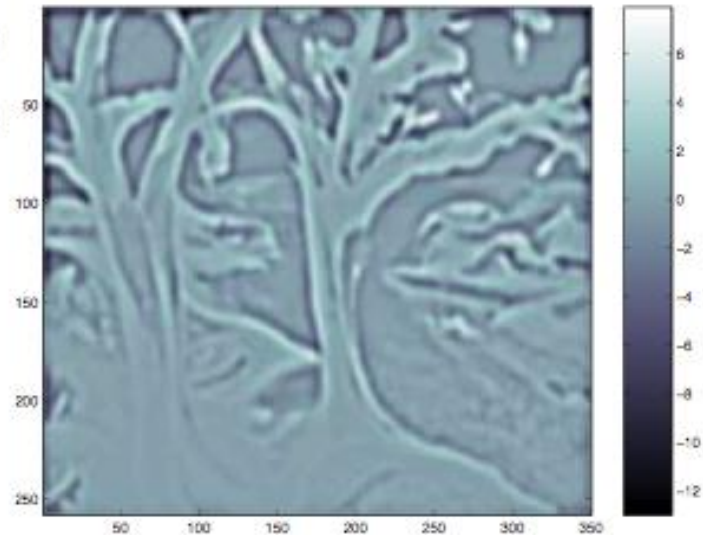


$$(w_{ij}) = \begin{pmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\Delta f = \sum_{i,j=-1,0,1} w_{ij} f(x_i, y_i)$$



Gaussian smoothing



Laplacian

Edge detection using sample variance

Often simple method performs far better than state-of-the-arts

Sample variance within moving window



Without smoothing

With smoothing

1 Diffusion equation in Euclidean space

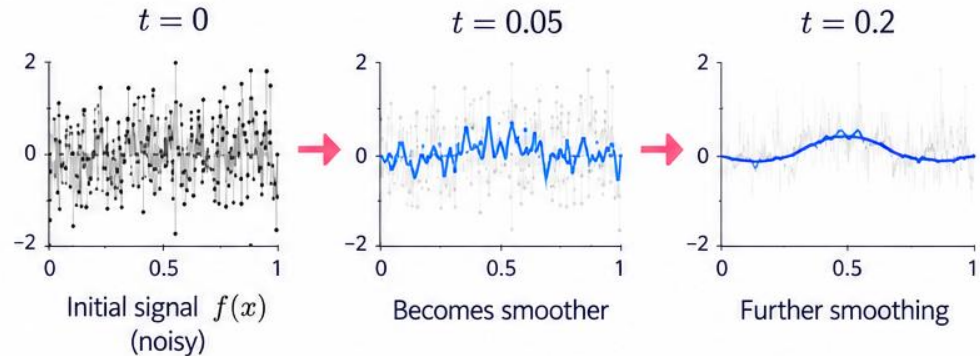
Heat (diffusion) equation:

$$\frac{\partial g(x, t)}{\partial t} = \Delta g(x, t) \quad x \in \mathbb{R}^n, \quad t > 0$$

Initial condition: $g(x, 0) = f(x)$

The function $g(x, t)$ (e.g., temperature or signal) spreads out and becomes smoother over time.

Example: diffusion of a noisy signal (1D)



2 Numerical solution by matrix multiplication

Discretize space on a grid with spacing h (1D):

$$\Delta g(x_i, t) \approx \frac{g_{i+1}(t) - 2g_i(t) + g_{i-1}(t)}{h^2}$$

Write in matrix form:

$$\frac{dg}{dt} = Lg, \quad g(0) = f$$

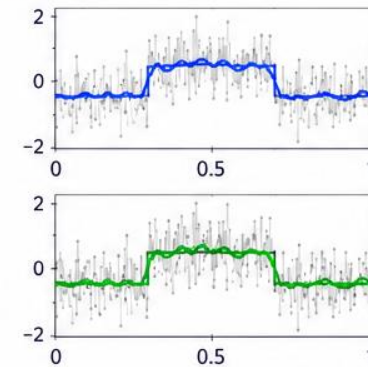
Solve by forward Euler iteration:

$$g^{n+1} = g^n + \Delta t L g^n = (I + \Delta t L) g^n$$

1D Laplacian matrix (N points)

$$L = \frac{1}{h^2} \begin{pmatrix} -2 & 1 & & & \\ 1 & -2 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & & 1 & -2 \end{pmatrix}$$

Result



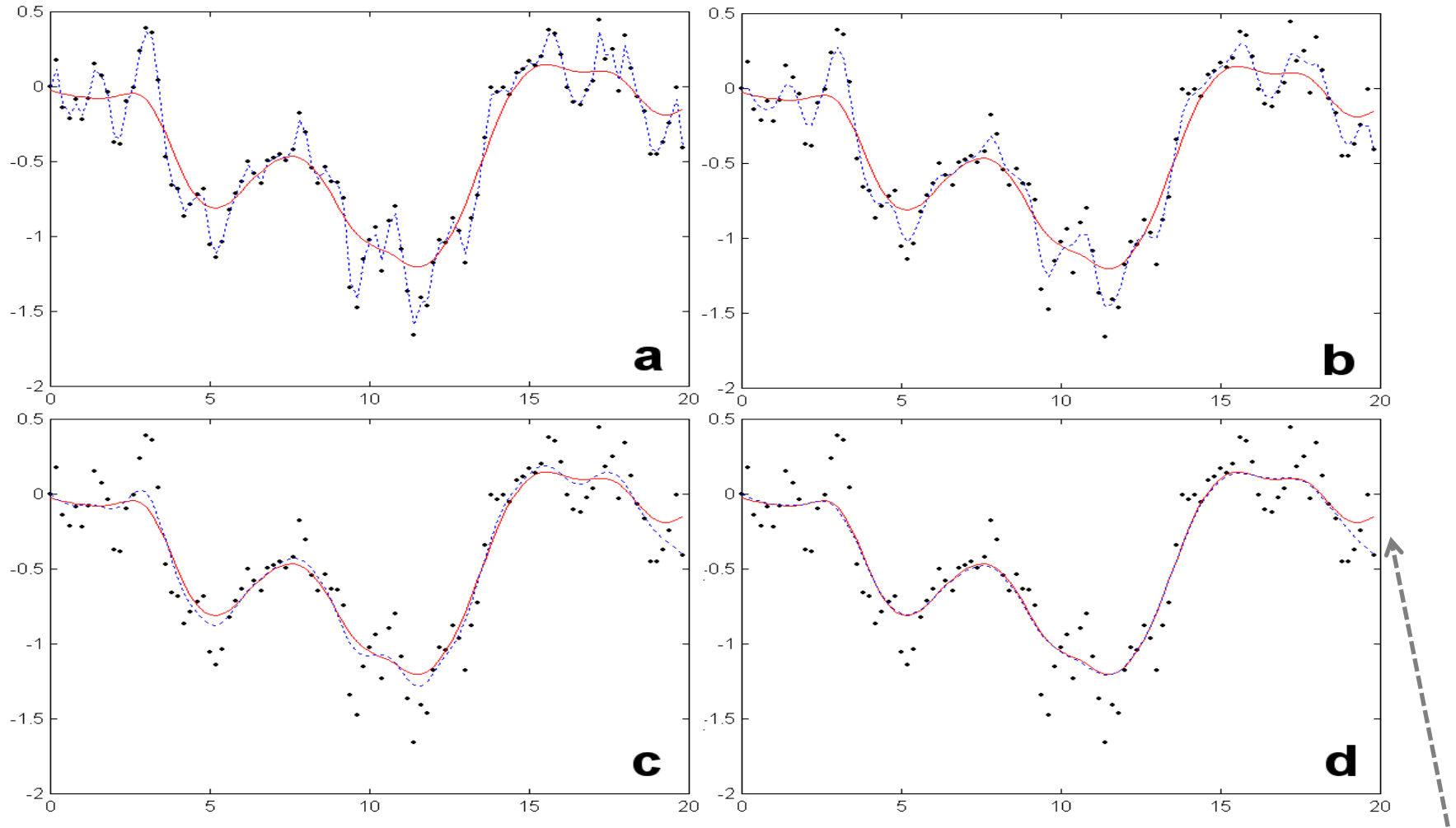
Iterative matrix multiplication result

Toeplitz matrix result

The results exactly match.

Diffusion smooths a function by repeatedly averaging with its neighbors.

Diffusion equation on 1D functional data



Red= Kernel density estimation

Blue = Diffusion equation after 5, 25 and 50 iterations

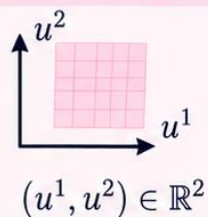
Shrinkage
at the
boundary

Laplace-Beltrami operator on manifold

1 Geometry and metric tensor

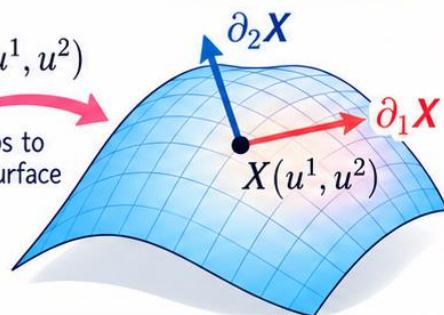
A 2D surface $\mathcal{M}$ is locally parameterized by coordinates (u^1, u^2) in $\mathbb{R}^2$.

Parameter domain



$X(u^1, u^2)$
maps to
the surface

Surface $\mathcal{M} \subset \mathbb{R}^3$

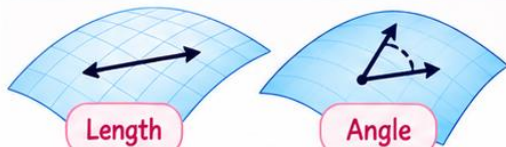


Metric tensor

The metric tensor measures lengths and angles on the surface.

$$g_{ij} = \langle \partial_i X, \partial_j X \rangle \quad (i, j = 1, 2)$$

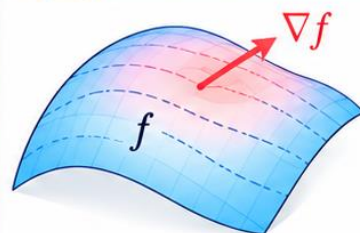
It encodes all geometric quantities (e.g., gradient, divergence, and Laplacian) intrinsically on the surface.



Lengths, angles, and differential operators are defined by the metric g .

2 Gradient, divergence, and Laplace-Beltrami

2.1 Gradient of a scalar function

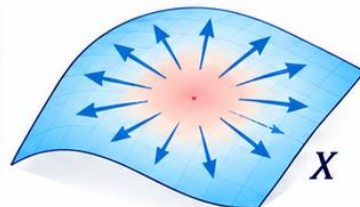


∇f is the tangent vector field of f , pointing in the direction of the steepest increase on the surface.

$$(\nabla f)^i = g^{ij} \partial_j f$$

∇f : gradient of f
 g^{ij} : inverse of the metric
(g^{ij} is the inverse of g_{ij})
 $\partial_j f$: partial derivative

2.2 Divergence of a vector field

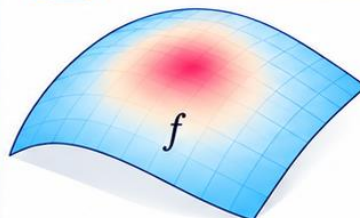


$\nabla \cdot X$ measures how much the vector field X spreads out (from a point) on the surface.

$$\nabla \cdot X = \frac{1}{\sqrt{|g|}} \partial_i (\sqrt{|g|} X^i)$$

$X = X^i \partial_i$: a tangent vector field
 $|g| = \det(g_{ij})$

2.3 Laplace-Beltrami operator



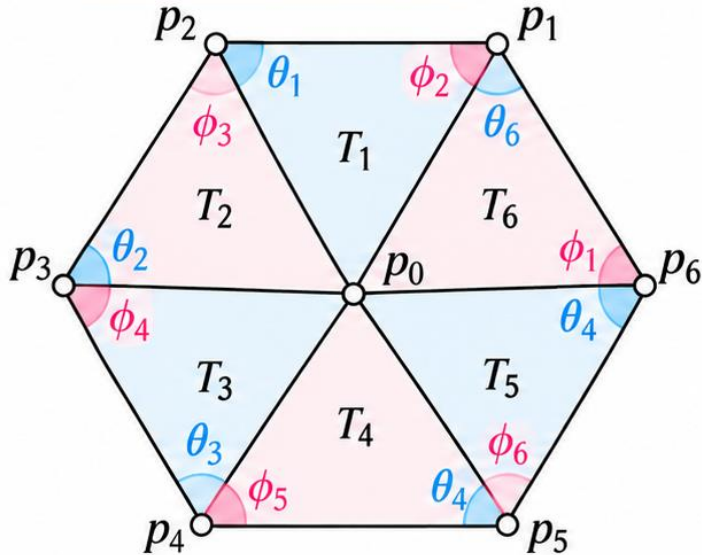
Δf measures how f differs from the average of its neighbors on the surface. It is the intrinsic diffusion operator.

$$\begin{aligned} \Delta f &= \nabla \cdot (\nabla f) \\ &= \frac{1}{\sqrt{|g|}} \partial_i (\sqrt{|g|} g^{ij} \partial_j f) \end{aligned}$$

Δ : Laplace-Beltrami operator
(= diffusion operator on the surface)

Cotan discretization of Laplace operator over 2-simplices

1 One-ring around p_0

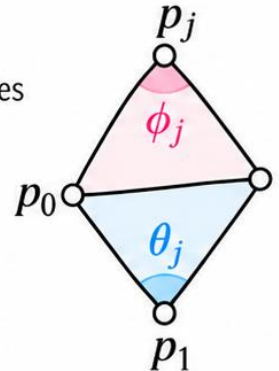


The triangles $T_1, \dots, T_6$ form the one-ring neighborhood of p_0 .

2 Cotan weight

For each edge (p_0, p_j) , let θ_j and ϕ_j be the two angles opposite to the edge (see the adjacent triangles).

$$w_j = \frac{\cot \theta_j + \cot \phi_j}{\sum_{i=1}^m |T_i|}$$



3 Discrete Laplace operator

The cotan discretization of the Laplace operator at vertex p_0 is

$$\Delta f(p_0) = \sum_{i=1}^m w_i [f(p_i) - f(p_0)]$$

4 Key points

- ✓ Uses only local 1-ring neighbors
- ✓ Weights are based on triangle angles (cotangent weights)
- ✓ Widely used in geometry processing and computer graphics

Local geometry $\rightarrow$ Laplace operator

Chung et al. 2003 NeuroImage
Chung et al. 2003 CVPR

Proof: Lopez-Perez et al, 2004, ECCV



diffusion on mesh

Unit Outcomes

- 1) Understand differential operators (DO)
- 2) Learn how to solve DO mathematically
- 3) Learn how to solve DO numerically
- 4) Learn how to set up diffusion equation as matrix multiplication